

SENTIMENT ANALYSIS USING NLP
MODEL EVALUATION REPORT
INTERSHIP TASK 02 – AI / ML

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1. Evaluation Overview:

After completing the sentiment analysis model, I tested it on the IMDb test dataset to see how well it could classify movie reviews. The model was tested on **25,000 reviews** that were not used during training. I used four evaluation metrics to check the results: Accuracy, Precision, Recall, and F1-Score. The purpose of this evaluation was to check how many reviews the model classified correctly and how well it performed for both positive and negative sentiments.

2. Model Used:

For sentiment classification, I used **Logistic Regression**. The model was trained using the TF-IDF features prepared from the IMDb reviews. After training, it was used to predict the sentiment of the test reviews. I used Logistic Regression because it is a simple classification model and works well with text features such as TF-IDF.

3. Test Dataset:

The evaluation was performed on **25,000 IMDb test reviews**.

The test dataset contains:

- **12,500 Negative reviews**
- **12,500 Positive reviews**

The test data was kept separate from the training data, so the evaluation shows how the model performed on reviews it had not seen during training.

4. Evaluation Metrics:

After training the model, I used four evaluation metrics to check its performance. These metrics were Accuracy, Precision, Recall, and F1-Score.

Accuracy shows how many predictions were correct out of all the test reviews.

Precision shows how many of the reviews predicted as positive were actually positive.

Recall shows how many of the actual positive reviews were correctly identified by the model.

F1-Score combines Precision and Recall and gives a balanced view of the model's performance.

5. Model Evaluation:

To check the performance of the model, I used the four evaluation metrics required for the task:

- Accuracy
- Precision
- Recall
- F1-Score

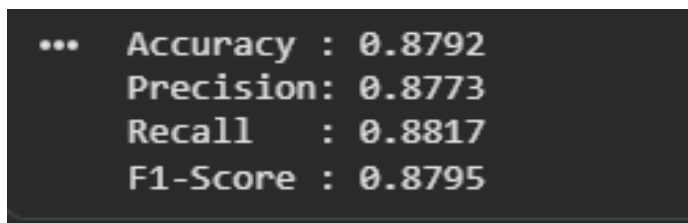
The model produced the following results:

Accuracy: 87.92%

Precision: 87.73%

Recall: 88.17%

F1-Score: 87.95%



```
... Accuracy : 0.8792
    Precision: 0.8773
    Recall   : 0.8817
    F1-Score  : 0.8795
```

The model achieved 87.92% accuracy, which means that most of the test reviews were classified correctly. The other scores were also close to 88%, which shows that the model performed fairly well overall.

6. Classification Report:

The classification report gives the performance of the model for both negative and positive reviews.

For **negative reviews**, the precision, recall, and F1-Score were all around **0.88**, with **12,500 reviews** in this class.

For **positive reviews**, the precision, recall, and F1-Score were also around **0.88**, with **12,500 reviews**.

The results for both classes are very similar. This shows that the model performed almost equally well when classifying positive and negative reviews.

...		precision	recall	f1-score	support
	Negative	0.88	0.88	0.88	12500
	Positive	0.88	0.88	0.88	12500
	accuracy			0.88	25000
	macro avg	0.88	0.88	0.88	25000
	weighted avg	0.88	0.88	0.88	25000

7. Confusion Matrix

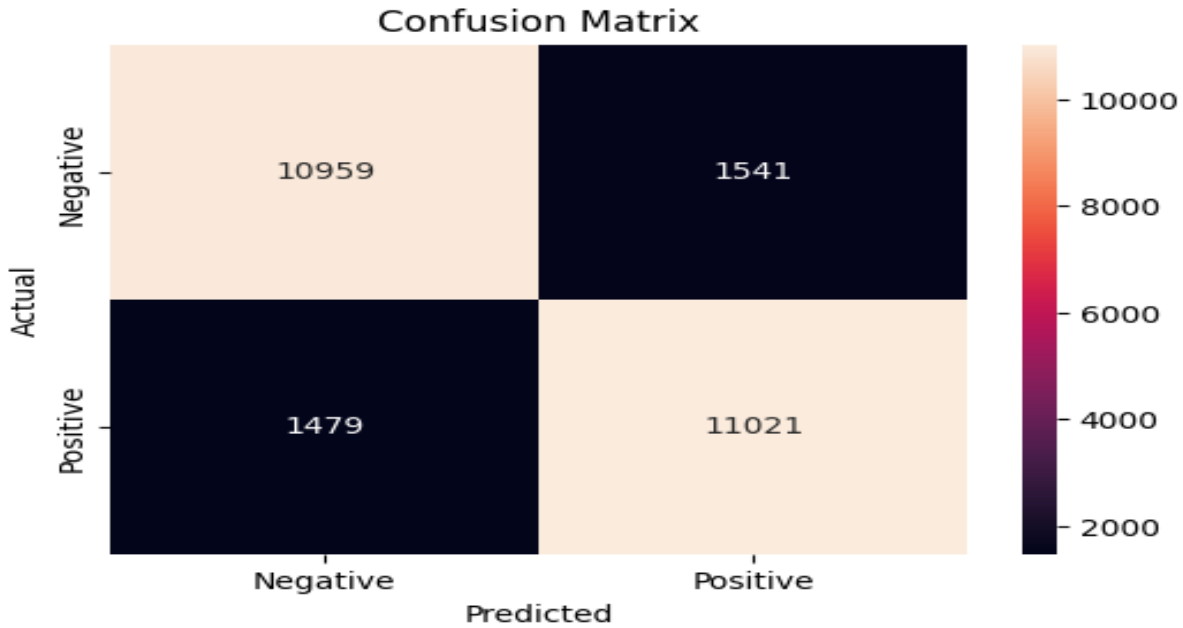
The confusion matrix obtained from the model was:

```
... [[10959 1541]
     [ 1479 11021]]
```

The results can be understood as follows:

- **10,959** negative reviews were correctly predicted as negative.
- **1,541** negative reviews were predicted as positive.
- **1,479** positive reviews were predicted as negative.
- **11,021** positive reviews were correctly predicted as positive.

So, the model correctly classified **21,980 out of 25,000** test reviews.



8. Confusion Matrix Analysis:

Looking at the confusion matrix, the model performed quite similarly for both positive and negative reviews. The number of false positive predictions was **1,541**, while the number of false negative predictions was **1,479**. These numbers are fairly close to each other. This means that the model did not strongly favor one sentiment class over the other. Overall, the number of correct predictions was much higher than the incorrect predictions.

9. Prediction Analysis:

I also checked some individual reviews from the test dataset to see how the model was making predictions. For these examples, I compared the actual sentiment with the sentiment predicted by the model. The model was able to correctly classify many of the reviews. However, some reviews can still be difficult to classify. For example, a review may contain both positive and negative opinions, sarcasm, or words that are difficult to understand based only on their frequency. Checking individual predictions helped me understand how the model performs on actual review text.

10. Overall Performance Analysis:

Overall, the model gave a good performance on the IMDb test dataset. The model achieved **87.92% accuracy**, while the precision was **87.73%**, recall was **88.17%**, and the F1-Score was **87.95%**. All four scores are close to each other, which shows that the model performed fairly well on both positive and negative reviews. The confusion matrix also supports these results because the model correctly classified a large number of reviews. For a basic sentiment analysis approach using TF-IDF and Logistic Regression, I think the results are good. However, there is still room for improvement.

11. Strengths of the Model:

Based on the evaluation results, I found the following strengths of the model:

- The model achieved **87.92% accuracy**.
- It performed well on both positive and negative reviews.
- Precision and Recall were both close to 88%.
- The F1-Score was also close to the accuracy score.
- The model was able to handle a test dataset of **25,000 reviews**.
- The number of correct predictions was much higher than the incorrect predictions.

12. Limitations:

The model also has some limitations. One limitation is that TF-IDF mainly focuses on the importance of words. It does not fully understand the meaning or context of a sentence. Because of this, the model may have difficulty with sarcasm, mixed opinions, or unusual wording. Another limitation is that the model was tested on IMDb movie reviews only. Its performance may be different when it is used on other types of text, such as tweets, customer feedback, or social media comments.

13. Possible Improvements:

There are several ways to improve this project in the future.

- Try other machine learning models and compare their results.
- Adjust the parameters of the Logistic Regression model.
- Experiment with different TF-IDF settings.
- Try word embeddings for better text representation.
- Use advanced NLP models such as BERT.
- Test the model on other datasets.
- Compare different models to find the best performing one.

- Improve the handling of sarcasm and sentence context.
- These improvements could help the model perform better on more difficult text

14. Practical Applications:

Sentiment analysis can be used in many real-world situations.

- **Customer Feedback:**
Companies can analyze customer feedback to find out whether customers are satisfied or dissatisfied.
- **Product Reviews:**
Online stores can use sentiment analysis to analyze large numbers of product reviews and understand customer opinions.
- **Marketing:**
Marketing teams can analyze customer responses to understand how people react to products or campaigns.
- **Customer Support:**
Customer support teams can identify negative feedback and focus on complaints that need attention.
- **Social Media Feedback:**
Sentiment analysis can also be used to understand people's opinions and reactions on social media.

15. Conclusion:

In this project, I evaluated a sentiment analysis model using IMDb movie reviews. The model used TF-IDF features with Logistic Regression to classify reviews as positive or negative. The model achieved **87.92% accuracy, 87.73% precision, 88.17% recall, and 87.95% F1-Score** on the test dataset. The results show that the model was able to classify most of the reviews correctly and performed fairly well for both positive and negative sentiments. There were still some incorrect predictions, especially in reviews that may contain mixed opinions, sarcasm, or difficult wording. However, the overall performance was good for the approach used in this project. This project helped me understand how to evaluate a machine learning model using different metrics and how to analyze the results to find the strengths and limitations of the model.

